

13 A For water, $(\Delta\theta_{\text{water}})_{\text{expt B}} = 3 (\Delta\theta_{\text{water}})_{\text{expt A}}$

Gain in heat, $Q_{\text{water}} \propto \Delta\theta_{\text{water}} \Rightarrow (Q_{\text{water}})_{\text{expt B}} = 3 (Q_{\text{water}})_{\text{expt A}}$

$\therefore$ Loss in heat for objects to water, $Q_B = 3 Q_A$

For objects, $c \propto \frac{Q}{\Delta\theta}$

$$\frac{c_A}{c_B} = \frac{Q_A}{Q_B} \frac{\Delta\theta_B}{\Delta\theta_A} = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

14 D $B = 1/2 \langle c^2 \rangle$